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Combustion characteristics of partially decomposed $\text{NH}_3/\text{H}_2/\text{N}_2$ blends: Effects of equivalence ratio, temperature, and NH_3 decomposition ratio

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ABSTRACT

NH_3 partial cracking into $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures has emerged as a flexible strategy to improve fuel reactivity without external H_2 supply, yet their combustion characteristics under well-defined thermal conditions remain unclear. In this study, gaseous species evolution and combustion efficiency during $\text{NH}_3/\text{H}_2/\text{N}_2$ combustion were systematically investigated over wide ranges of equivalence ratio (0.4–1.4), temperature (700–1200 °C), and NH_3 decomposition ratio (0–0.7), with data reliability verified by nitrogen balance analysis. Increasing temperature progressively alleviated kinetic limitations and promoted combustion. N_2O formation was favored under lean-to-stoichiometric conditions at 700–800 °C and gradually decreased with temperature. In contrast, the temperature dependence of NO under lean conditions was governed by competing effects of enhanced radical production and NH_3 pre-decomposition. Combustion efficiency increased with temperature and plateaued under lean conditions, while slightly decreasing at high temperatures under rich conditions. Hydrogen blending markedly enhanced combustion efficiency, particularly at 700 °C and under rich conditions. The effects of NH_3 decomposition ratio on NO and N_2O formation were governed by the competition between enhanced reactive radical generation and the dilution of nitrogen-containing intermediates. The preferred operating condition depended on the target optimization objective. NH_3 combustion at 800–900 °C under slightly lean conditions provided a favorable balance between combustion efficiency and NOx emissions, whereas increasing NH_3 decomposition ratio further suppressed N_2O at the expense of higher NO formation. Temperatures above 1000 °C eliminated N_2O but caused substantial NO formation, while slightly rich conditions favored lower NO emissions with modest penalties in N_2O emissions and combustion efficiency at high decomposition ratio (i.e., 0.7). These findings offer practical guidance for optimizing combustion performance and emission characteristics, contributing to the development of decarbonized energy systems.

1. Introduction

With continued economic growth and industrial development, global energy-related CO_2 emissions increased by 0.8% in 2024, reaching a record 37.8 Gt CO_2 and driving atmospheric CO_2 concentrations to 422.5 ppm [1]. Such unprecedented CO_2 emissions highlight the urgent need for a low-carbon transition in the energy sector. In this context, ammonia has attracted growing attention as a promising carbon-free fuel that can be synthesized from renewable H_2 (via water electrolysis) and N_2 (via air separation) through a renewable energy-powered Haber-Bosch process. As an effective hydrogen carrier (17.7 wt%) [2], it benefits from well-established production, storage, and distribution

infrastructures, as well as easier liquefaction and lower storage and transportation costs compared with hydrogen [3,4]. Furthermore, its higher ignition energy and narrower flammability limits alleviate safety concerns by reducing explosion and flashback risks [5]. Nevertheless, ammonia combustion suffers from inherently poor reactivity, as reflected by its low laminar burning velocity, narrow flammability limits, low adiabatic flame temperature, and long ignition delay time, posing challenges for stable and efficient combustion [6,7].

To overcome these limitations, blending ammonia with more reactive fuels, such as hydrogen [8,9], methane [10,11], CO [12,13], and dimethyl ether [14,15], has been widely proposed as an effective strategy to enhance fuel reactivity and flame stability. Among them,

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hydrogen blending is particularly attractive because it significantly increases laminar burning velocity while avoiding CO₂ emissions during co-combustion. Valera-Medina et al. [16,17] reported stable NH₃/H₂ premixed combustion under both lean and rich conditions, with enhanced NO_x formation observed under lean conditions. In contrast, Zhang et al. [18] observed improved stability without significant NO_x increase at low hydrogen fractions, highlight a strong dependence of NO_x formation on the H₂ blending ratio. In addition, bluff body burners have been shown to enhance NH₃/H₂ flame stability via the establishment of recirculation zones, extending stability limits at low hydrogen blending ratios without increasing flashback risk [19,20].

Despite the improvements in flame stability, NO_x emissions remain a critical challenge in ammonia combustion due to its inherently high nitrogen content. Various strategies have been proposed to mitigate NO_x formation in NH₃/H₂ system, including steam dilution, staged combustion, and novel burner design. Steam addition has been demonstrated to effectively reduce NO_x without compromising combustion efficiency, primarily by lowering flame temperature and scavenging O radicals [21]. To further clarify the relative roles of thermal, chemical, and transport effects of H₂O dilution, the fictitious gas method has been employed [22]. Rich-lean two-stage combustion has been widely investigated, where NO_x reduction generally benefits from longer residence times, elevated pressure and richer conditions in the primary zone [23], while the optimal primary equivalence ratio decreased exponentially with decreasing hydrogen blending ratios [24]. The combined application of elevated pressure, steam addition, and staged combustion further suppressed NO_x formation when the secondary air flow was properly controlled [25]. Moreover, radial two-stage combustion has been proposed as an alternative to conventional axial configurations, with a hydrogen-doped rich-premixed ammonia core surrounded by lean-premixed pure hydrogen flames, which has been demonstrated to enhance NO_x mitigation [26].

However, previous studies have predominantly relied on flame-based configurations, such as swirl burners, where the temperature field is intrinsically governed by strongly coupled interactions among chemical reactions, turbulent transport, and heat release, resulting in spatially and temporally evolving flame structures. In contrast, reactor-based systems, such as jet-stirred reactors (JSRs) and plug flow reactors (PFRs), provide externally controlled thermal environments, in which temperature can be independently regulated from local reaction dynamics. Such configurations are particularly valuable for investigating combustion, as they enable the analysis of intrinsic chemical kinetics and reaction pathways with minimized influence of coupled transport-reaction effects. Studies conducted in JSRs have demonstrated that hydrogen addition markedly lowers the onset temperature for NH₃ conversion. Meanwhile, NO emissions exhibit a non-monotonic dependence on the hydrogen blending ratio, reflecting the competition effects between reduced nitrogen-containing intermediates and H₂-induced chemical effects [27,28]. Furthermore, steam has been reported to exert opposite effects on NO formation under different equivalence ratios, suppressing NO under fuel-lean conditions while promoting it under fuel-rich conditions [29]. In addition, mutual inhibitory interactions between NH₃ and H₂ has been reported in JSR experiments [30]. Beyond JSR investigations, flow reactor has also been employed to investigate the oxidation characteristics of NH₃/H₂ combustion. H₂ addition progressively enhanced fuel reactivity [31], with the extent of enhancement being approximately proportional to the H₂ mole fraction [32]. Moreover, both elevated pressure and higher H₂/NH₃ ratios shifted the onset of fuel conversion towards low temperatures, while simultaneously promoting N₂O formation.

Despite these extensive efforts, most existing studies have focused on simplified NH₃/H₂ fuel mixtures, which implicitly assume an independent external H₂ supply. However, such a strategy is difficult to implement in practical combustion systems due to the challenges associated with hydrogen storage, transportation, infrastructure compatibility, and on-site safety management. Beyond these constraints, external hydrogen

supply would require an additional independent fuel delivery system, significantly increasing the complexity and cost of ammonia combustion systems. Under such circumstances, partial ammonia cracking has become a practically unavoidable strategy as it enables in situ H₂ generation directly from NH₃ fuel without requiring external hydrogen supply [33]. However, practical NH₃ cracking inevitably produces NH₃/H₂/N₂ mixtures rather than simplified NH₃/H₂ mixtures, since N₂ is an intrinsic product of NH₃ decomposition. Consequently, NH₃/H₂/N₂ mixtures are more representative of practical ammonia combustion system with in-situ cracking. Moreover, the presence of N₂ can further modify combustion characteristics compared with NH₃/H₂ mixtures. In terms of flame stability, studies on NH₃/H₂/N₂ flames have shown that the introduction of N₂ did not significantly alter the flame stability limits [9]. Ji et al. [34] further demonstrated that NH₃/H₂/N₂ mixtures exhibited lower laminar burning velocities than the corresponding NH₃/H₂ mixtures, and that the inhibitory effects of N₂ became increasingly pronounced with increasing NH₃ decomposition ratio. Beyond flame stability, N₂ also affects pollutant formation. Okumura et al. [35] reported that N₂ introduced through NH₃ cracking reduced flame temperature and OH concentration, resulting in lower NO emissions. Consistent NO reduction behavior was also observed when investigating the dilution effect of N₂ on NH₃/H₂/air combustion [36]. In addition, numerical simulations by Sheykhabaglou et al., [37] revealed that partially cracked NH₃ flames exhibited flow characteristics (i.e., recirculation patterns) distinct from those of NH₃/H₂ flames, suggesting distinct flame stabilization mechanisms and pollution formation pathways. However, despite the increasing number of studies on NH₃/H₂/N₂ flames, experimental investigations of partially cracked NH₃/H₂/N₂ mixtures in thermally controlled reactors remain limited, particularly under varying equivalence ratios, temperatures, and NH₃ decomposition ratios. This lack of comprehensive understanding hinders the optimization and implementation of practical ammonia combustion systems.

Therefore, this study aims to evaluate the combustion performance of partially cracked NH₃/H₂/N₂ mixtures in an externally heated, temperature-controlled tubular flow reactor, addressing the lack of data under well-defined thermal conditions. The effects of equivalence ratio, temperature, and NH₃ decomposition ratio on gaseous species evolution and combustion efficiency were systematically investigated and comprehensively evaluated. The results substantially expand the experimental database for NH₃/H₂/N₂ combustion under externally controlled thermal conditions and provide actionable guidance for the effective utilization of ammonia-based fuels in practical combustion systems, thereby supporting the development of near-term decarbonization strategies in the energy sector.

2. Method

2.1. Experimental setup

A schematic diagram of the experimental setup is illustrated in Fig. 1. The system consisted of three main components: a gas feeding unit, a high-temperature reaction system, and a gas analysis system. All reactant gases were supplied from compressed gas cylinders. The gas flow rates were regulated by rotameters and continuously monitored by mass flow meters to ensure precise control. Depending on the target flow rate of each gas, either KOFLOC flow meters (KOFLOC Corp., Japan; full-scale range: 5–20 L/min) or Alicat flow meters (Alicat Scientific, Japan; full-scale range: 100 mL/min) were employed. Downstream of the flow control system, the gases were divided into two separate streams to prevent reactions prior to entering the isothermal zone. The fuel stream, consisting of NH₃, H₂ and N₂, was designated to simulate partially decomposed ammonia, and was introduced at the onset of the isothermal zone through a mullite fuel lance (inner diameter: 11 mm). The tip of the lance was sealed, and fuel was injected radially through two circumferential rows of sidewall orifices, with five evenly distributed holes in each row. This configuration was designed to enhance

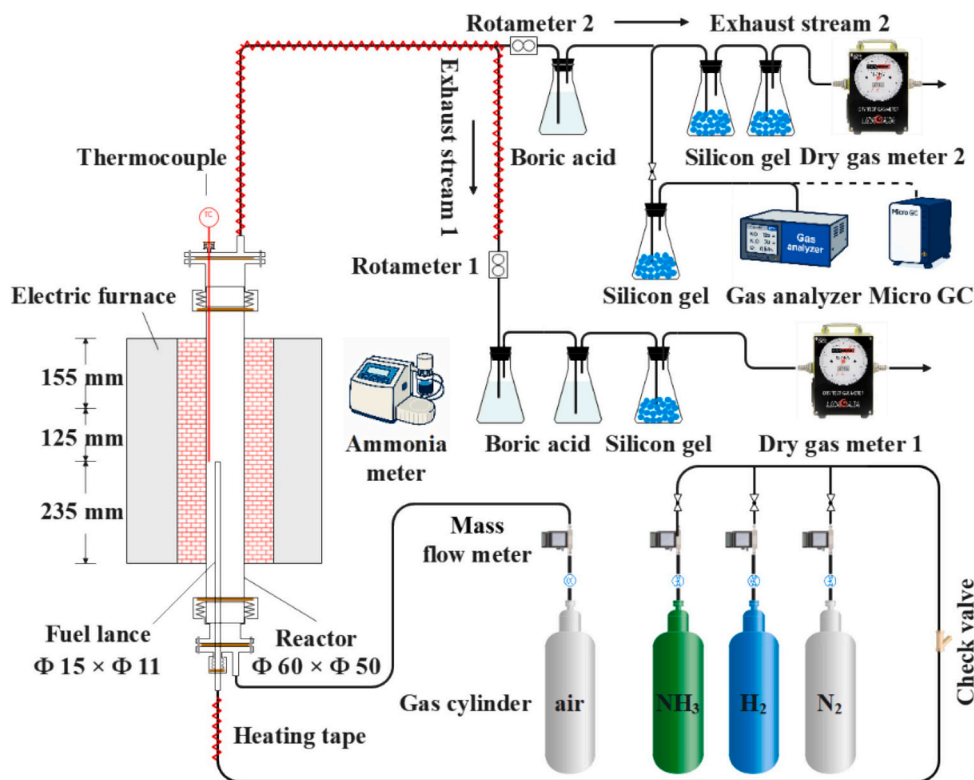


Fig. 1. Schematic diagram of the experimental setup.

mixing between fuel and the surrounding oxidizer. The oxidizer stream, composed of air, was introduced from the bottom of the reactor through a gas inlet nozzle.

Combustion experiments were conducted in a vertically mounted alumina reactor (inner diameter: 50 mm; length: 1000 mm) housed inside an electric tube furnace (Carbolite Gero, Germany). The furnace had a maximum operating temperature of 1800 °C. Based on axial temperature distribution profiles measured using a K-type thermocouple (Fig. 2), the isothermal zone was determined to have a length of approximately 125 mm, with its lower boundary located 235 mm above the bottom of the furnace. A discrepancy was observed between the set temperatures and the measured gas temperatures in the isothermal zone.

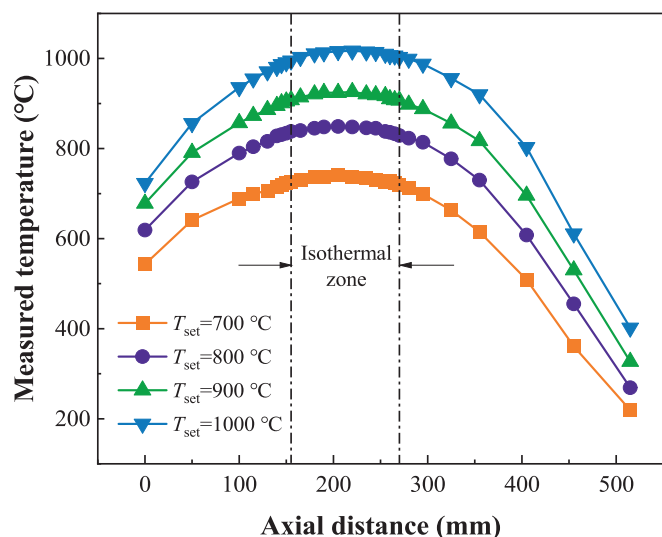


Fig. 2. Axial temperature distribution along the reactor, with the furnace top defined as the axial origin.

While the deviation was within 17 °C at a set temperature of 1000 °C, it became more pronounced at lower set temperatures. To account for the discrepancy, the temperatures presented in the subsequent figures correspond to the measured pre-reaction temperatures at the entrance of the isothermal zone rather than the set temperatures. As the discrepancy between the set and measured temperatures had been corrected by adopting the measured pre-reaction temperatures, the remaining axial temperature variation within the isothermal zone (< 20 °C) was considered to have a negligible influence on the observed experimental trends. Given that minor variations in the measured pre-reaction temperatures existed among individual runs within the same set temperature group, these measured values were not treated as independent experimental conditions. Instead, the set temperatures were retained throughout the main text to designate the different experimental groups and facilitate comparisons among them.

To suppress water condensation and NH_3 dissolution, the exhaust line was maintained at 120 °C using heating tapes. The exhaust gas was subsequently divided into two streams with flow rates controlled by rotameters. One stream (exhaust stream 1) was directed through boric acid absorption bottles for residual NH_3 capture, followed by moisture removal using silica gel, and the cumulative gas volume was measured with a dry gas meter (Shinagawa Co., Ltd., Japan). The other stream (exhaust stream 2) was used for ventilation and gas analysis. After NH_3 removal by boric acid solution, it was further divided into a ventilation sub-stream monitored by a dry gas meter, and an analytical sub-stream for gas composition analysis using either an online gas analyzer (VA-5114, Horiba, Japan) or a micro gas chromatograph (Micro GC, Agilent, USA). For online analysis, the gas was measured directly after moisture removal, whereas for Micro GC analysis, the gas was first dried and collected in gas sampling bags prior to measurement.

2.2. Procedures

Prior to experiments, the coulometric ammonia analyzer (Quick Ammonia, Central Kagaku Corp., Japan), VA-5114 gas analyzer, and

Micro GC were calibrated using standard solution or calibration gases. Before each run, both rotameters 1 and 2 were fully opened, and the analytical sub-stream remained closed. The boric acid absorption unit intended for connection to rotameter 1 was disconnected from the flow line and the initial reading of dry gas meter 1 ($V_{1,i}$, L) was recorded.

After the furnace reached the target temperature, the inlet gas flow rates were adjusted to their preset values. Rotameter 2 was then regulated so that rotameter 1 maintained a reading of approximately 70 mL/min, ensuring complete NH_3 absorption. After the system had stabilized for approximately 10 min, the initial cumulative gas volumes indicated by the mass flow meters ($V_{i,\text{air}}$, V_{i,NH_3} , V_{i,H_2} , and V_{i,N_2} , L) and the corresponding time (t_i , min) were recorded. Subsequently, the boric acid absorption unit was connected to rotameter 1, and the variations in the readings of the two dry gas meters over a 10-min interval were recorded to determine the total exhaust gas flow rate ($\dot{V}_{\text{out}}$, L/min).

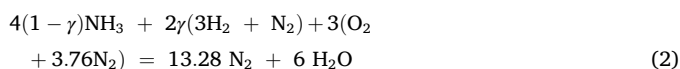
The valve of the analytical sub-stream was then opened, and the exhaust gas was directly introduced into the gas analyzer for online NO and N_2O measurements. After NOx analysis, the exhaust gas was collected in a sampling bag and subsequently analyzed using the Micro GC. Finally, the boric acid absorption unit was again disconnected from rotameter 1, and the final cumulative volumes recorded by mass flow meters ($V_{f,\text{air}}$, V_{f,NH_3} , V_{f,H_2} , and V_{f,N_2} , L), together with the corresponding recording time (t_f , min), were logged to calculate the actual stoichiometric ratio. The final reading of the dry gas meter 1 ($V_{1,f}$, L) was also recorded. The boric acid solutions in the absorption unit were thoroughly mixed and collected for subsequent $\text{NH}_4^+\text{-N}$ analysis using the ammonia analyzer.

2.3. Experimental conditions

The effects of three operating parameters, namely equivalence ratio (Φ), temperature (T), and NH_3 decomposition ratio (γ), were systematically investigated. The equivalence ratio (Φ) is defined in Eq. (1), where m and V denote mass and volume, respectively, and the subscripts stoich. and act. refer to stoichiometric and actual combustion conditions.

$$\Phi = \frac{(m_{\text{fuel}}/m_{\text{air}})_{\text{act.}}}{(m_{\text{fuel}}/m_{\text{air}})_{\text{stoich.}}} = \frac{V_{\text{air, stoich.}}}{V_{\text{air, act.}}} \quad (1)$$

Under stoichiometric conditions, the global reaction can be expressed as follows:



Here, γ denotes the NH_3 decomposition ratio, defined as the fraction of ammonia decomposed via the reaction $2\text{NH}_3 = \text{N}_2 + 3\text{H}_2$ relative to the total ammonia supplied, including both decomposed and undecomposed fractions, as expressed in Eq. (3), where n denotes the number of moles.

$$\gamma = \frac{2 \times n_{\text{H}_2}}{2 \times n_{\text{H}_2} + 3 \times n_{\text{NH}_3}} \quad (3)$$

For all experimental cases, the thermal input power was maintained at 0.1 kW, calculated based on the volumetric energy densities of NH_3 (14.34 MJ/m³) and H_2 (10.68 MJ/m³) derived from their lower heating values and densities reported in the literature [38]. A parametric study was performed by systematically varying the equivalence ratio, temperature and NH_3 decomposition ratio. The detailed experimental conditions are listed in Table 1. The temperature range of 700–1200 °C was selected to span from the onset of NH_3 oxidation to the representative furnace temperatures of industrial gas-fired boilers (typically 1000–1400 °C). Temperatures above 1300 °C were not considered, as thermal NO formation becomes increasingly significant at elevated temperatures, which may obscure fuel-N reaction pathways. Meanwhile, temperatures below 700 °C were excluded due to reduced temperature

Table 1

Experimental conditions in the current work.

$T/^\circ\text{C}$	Φ	γ	Re^a	Re^b	$\chi_{\text{H}_2}^c$	τ^d/s
700	0.4, 0.95, 1.05, 1.4	0, 0.3	40.4–114.1	16.2–49.6	0, 0.39	1.08–3.03
800	0.4–1.4	0, 0.1, 0.3, 0.5, 0.7	39.2–114.1	14.2–46.3	0–0.78	0.98–2.74
900	0.4, 0.95, 1.05, 1.4	0, 0.3	40.4–114.1	14.1–43.6	0, 0.39	0.90–2.51
1000	0.4–1.4	0, 0.1, 0.3, 0.5, 0.7	39.2–114.1	12.7–41.2	0–0.78	0.83–2.31
1100	0.4, 0.95, 1.05, 1.4	0, 0.3	40.4–114.1	12.7–39.1	0, 0.39	0.77–2.14
1200	0.4, 0.95, 1.05, 1.4	0, 0.3	40.4–114.1	12.1–37.3	0, 0.39	0.71–2.00

^a Calculated at room temperature; ^b Calculated at reaction temperature; ^c Volume fraction of H_2 in the H_2/NH_3 mixture; ^d Residence time in isothermal zone.

reliability, as deviations between the set and actual gas-phase temperatures become more pronounced. The equivalence ratio was varied from 0.4 to 1.4 to cover conditions from fuel-lean to fuel-rich, while the NH_3 decomposition ratio ranged from 0 to 0.7. Higher NH_3 decomposition ratios were excluded, as the corresponding NH_3 flow rates would fall below 2% of the full-scale range, leading to increased measurement uncertainty. To provide a more intuitive description of the fuel composition, the H_2 molar ratio, χ_{H_2} , defined as the molar fraction of H_2 relative to the total moles of NH_3 and H_2 , is also reported in Table 1. The Reynolds number of the reactant gases in the alumina tube ranged from 12.1 to 49.6 under the actual reaction temperatures investigated in this study (corresponding to $\text{Re} = 39.2\text{--}114.1$ at room temperature). These values indicate laminar flow throughout the experiments, thereby minimizing turbulence-induced mixing effects on the measured reaction characteristics. The residence time within the isothermal zone ranged from 0.71 s to 3.03 s depending on the operating conditions. An approximate estimation of the characteristic mixing time suggests that jet-induced radial transport occurs on a timescale short compared with the gas residence time, indicating that radial mixing is unlikely to be a limiting factor under the present experimental conditions (see Supplemental Material for details). Each experimental condition was repeated three times, and the reported results correspond to the average of the three measurements. Error bars shown in the subsequent figures represent the standard deviation of the repeated measurements.

2.4. Data analysis

The volume concentration of NH_3 ($[\text{NH}_3]$, ppm) in the exhaust gas on a dry basis was calculated according to the following equation:

$$[\text{NH}_3] = \frac{c_{\text{NH}_4^+} \times V_{\text{sol}} \times 10^{-3} \times 22.4/18}{c_{\text{NH}_4^+} \times V_{\text{sol}} \times 10^{-3} \times 22.4/18 + (V_{1,f} - V_{1,i})} \times 10^6 \quad (4)$$

Where $c_{\text{NH}_4^+}$ represents the mass concentration of NH_4^+ in the collected boric solution (mg/L) and V_{sol} is the total volume of the boric acid solution used for exhaust stream 1 (L). The values of 22.4 and 18 correspond to standard molar volume of an ideal gas at STP and the molar mass of NH_4^+ , respectively.

To assess the reliability of the experimental data, a mass balance analysis was performed. Specifically, the nitrogen balance (%), defined as the ratio of total nitrogen atoms (on an elemental basis) measured in

the outlet gas to those in the inlet gas, was evaluated according to the following equation:

$$\text{N balance} = \frac{\dot{V}_{\text{out}} \times (2 \times x_{\text{N}_2} + x_{\text{NO}} + 2 \times x_{\text{N}_2\text{O}}) + V_{\text{O}, \text{NH}_3} / (t_f - t_i)}{(2 \times 0.79 \times (V_{\text{f, air}} - V_{\text{i, air}}) + 2 \times (V_{\text{f, N}_2} - V_{\text{i, N}_2}) + (V_{\text{f, NH}_3} - V_{\text{i, NH}_3})) / (t_f - t_i)} \times 100\% \quad (5)$$

Where x_{N_2} , x_{NO} , and $x_{\text{N}_2\text{O}}$ denote the volume fractions of N_2 , NO , and N_2O in the exhaust gas, respectively. The term $V_{\text{O}, \text{NH}_3}$ (L) represents the total amount of NH_3 in the exhaust over the measurement period ($t_f - t_i$, min). It was calculated based on $c_{\text{NH}_4^+}$ and V_{sol} , followed by a correction using the flow-rate ratio between exhaust streams 1 and 2, assuming complete absorption of NH_3 in the boric acid solution. NO_2 was not included in the N balance due to the lack of a dedicated NO_2 detection system. Nevertheless, nitrogen oxides contribute negligibly to the overall N balance compared with N_2 and NH_3 in the exhaust gases. Moreover, NO_2 concentration has been reported to be orders of magnitude lower than that of NO due to its thermodynamic instability [7,39]. Therefore, the omission of NO_2 introduces only negligible uncertainty in the N balance.

To facilitate comparison among different equivalence ratios, the measured NO and N_2O concentrations ($[\text{NO}]$ and $[\text{N}_2\text{O}]$, ppm) were corrected to a reference O_2 concentration of 3% on a dry basis, which is widely adopted in gas-fired boilers and gas turbines. Accordingly, the normalized concentrations were calculated using Eq. (6), where $[\text{O}_2]$ represents the volume concentration of O_2 in the exhaust gas, vol.%.

$$[\text{NO}/\text{N}_2\text{O}] \text{ at } 3\% \text{ O}_2 = [\text{NO}/\text{N}_2\text{O}] \times \frac{18}{21 - [\text{O}_2]} \quad (6)$$

To evaluate fuel utilization efficiency, both the energy-based combustion efficiency (η_e , %) and the mass-based (hydrogen-atom-based) combustion efficiency (η_m , %) were calculated. These efficiencies quantify the fractions of fuel's chemical energy and hydrogen atoms that are effectively consumed during combustion, excluding the portions remaining in the exhaust gas as unreacted NH_3 and H_2 , as defined in Eqs. (7)–(8).

$$\eta_e = \left[1 - \frac{\dot{V}_{\text{out}} \times x_{\text{H}_2} \times 10.68 + V_{\text{O}, \text{NH}_3} / (t_f - t_i) \times 14.34}{[(V_{\text{f, H}_2} - V_{\text{i, H}_2}) \times 10.68 + (V_{\text{f, NH}_3} - V_{\text{i, NH}_3}) \times 14.34] / (t_f - t_i)} \right] \times 100\% \quad (7)$$

$$\eta_m = \left[1 - \frac{\dot{V}_{\text{out}} \times x_{\text{H}_2} \times 2 + V_{\text{O}, \text{NH}_3} / (t_f - t_i) \times 3}{[(V_{\text{f, H}_2} - V_{\text{i, H}_2}) \times 2 + (V_{\text{f, NH}_3} - V_{\text{i, NH}_3}) \times 3] / (t_f - t_i)} \right] \times 100\% \quad (8)$$

Where x_{H_2} represents the volume fraction of H_2 in the exhaust gas.

3. Results and discussion

3.1. Mass balance analysis

The reliability of the experimental data was assessed through a nitrogen balance analysis. As shown in Fig. 3, the N balance ranges from 96.7% to 110.0%, indicating the overall reliability of the measurements when considering the inherent uncertainties associated with flow control and gas quantification. The N balance generally exhibits a non-monotonic dependence on equivalence ratio, increasing under lean conditions, reaching a maximum near stoichiometry, and subsequently decreasing under rich conditions. Such a systematic trend suggests that the observed deviations are not governed by random experimental uncertainties, but rather reflect the shifting partitioning of nitrogen among N_2 and NH_3 as the equivalence ratio varies.

Under lean conditions, the nitrogen balance is primarily governed by N_2 , as the contributions from NO , N_2O , and NH_3 are comparatively minor due to their low concentrations, which will be discussed in detail in the following sections. The observed increasing trend in the nitrogen balance with equivalence ratio can be partially attributed to the over-estimation of the O_2 concentration, as the O_2 calibration of Micro GC was performed using air while assuming an O_2 fraction of 21%, which consequently leads to an underestimation of the N_2 concentration calculated by difference. As the equivalence ratio increases, the relative error associated with O_2 decreases due to lower O_2 concentration, thereby alleviating the underestimation of N_2 and resulting in a gradual increase in the calculated nitrogen balance. In contrast, under rich conditions, the contribution of NH_3 becomes significant. The observed

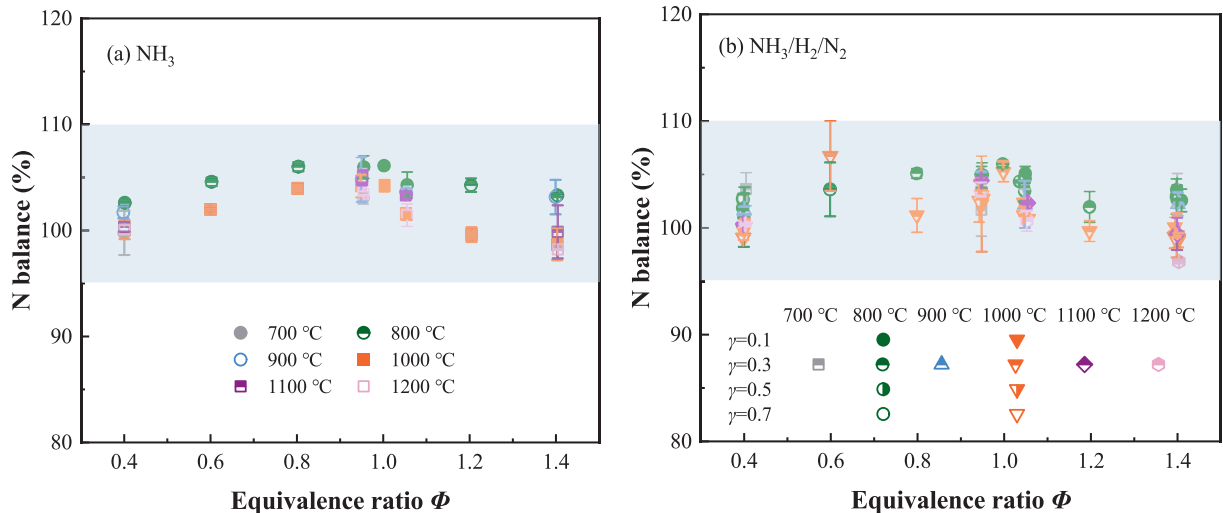


Fig. 3. Nitrogen balance analysis for (a) pure NH_3 and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends.

decrease in the nitrogen balance may be partially attributed to the underestimation of NH_3 concentration, potentially caused by adsorption losses on the pipeline surfaces. This effect becomes more pronounced at higher equivalence ratios, where increased NH_3 concentrations promote greater adsorption losses.

While the above discussion explains the observed dependence of the nitrogen balance on equivalence ratio, it cannot fully explain the deviation of the N balance from the ideal value of 100%. Such deviations are expected to arise from the accumulation of uncertainties associated with multiple measurements involved in the N balance calculation. Specifically, the major contributions include the flow meter accuracy (1.0%), dry gas meter repeatability (0.3%), and the mean relative errors of the O_2 , H_2 , and NH_3 measurements derived from calibration results (2.2%, 3.0% and 2.7%, respectively). Since the outlet N_2 concentration was determined by difference, uncertainties associated with O_2 and H_2 propagate into the calculated N_2 concentration and are further amplified by the coefficient of 2 in the N balance equation. Based on a root-sum-square estimation, the overall uncertainty of the N balance was estimated to be up to 7.9%, which is sufficient to account for the majority of the observed deviations from the ideal value of 100%, except for an isolated operating condition exhibiting an N balance of approximately 110%. These analyses support the overall reliability of the experimental dataset.

3.2. Effects of equivalence ratio

The evolutions of species concentration and combustion efficiency as a function of equivalence ratio at the preset temperatures of 800 °C and 1000 °C are depicted in Figs. 4 and 5, respectively. As the equivalence ratio increased, the NO concentration exhibited a non-monotonic trend, initially increasing, followed by a decrease, and eventually approaching a plateau under fuel-rich conditions ($\Phi > 1$). The decrease in NO concentration observed under ultra-lean conditions was likely attributable to volumetric dilution caused by the introduction of a large excess of air. To eliminate this dilution effect, the measured NO concentrations were corrected to a reference oxygen concentration of 3%. After O_2 normalization, the NO concentration decreased monotonically with increasing equivalence ratios and eventually approaching a plateau, suggesting a higher tendency for NO formation under lower equivalence ratios. Under fuel-lean conditions, abundant oxygen-containing radicals (e.g., OH, O and HO_2) facilitated the oxidation of nitrogen-containing intermediates (NH_2 , NH, etc.) to NO (R1-R5), whereas fuel-rich conditions favored the conversion of NH_i species into N_2 via both recombination reactions (R6-R7) and thermal DeNOx pathways ($\text{NH}_i + \text{NO}$, R8-R10) [4,7,39,40]. Notably, as shown in Fig. 4(b) and 5(a), hydrogen blending and temperature variation did not alter the overall NO variation trend with equivalence ratio.



Compared with NO, both the measured and O_2 -normalized N_2O concentration remained at much lower levels under lean conditions at 800 °C, showing only a weak dependence on equivalence ratio. As the equivalence ratio approached 1, the N_2O concentration increased slightly, reaching a maximum at $\Phi = 1$, and then gradually decreased to 0 under fuel-rich conditions. It has been reported that N_2O is mainly formed through $\text{NH} + \text{NO}$ (R9) [41] and $\text{NH}_2 + \text{NO}_2$ (R11) [40,42] reaction pathways, which compete with the oxidation of NH_i to NO via reactions with O, OH, and HO_2 radicals (R1-R5). A similar competition between N_2O -formation and NO-formation pathways has also been reported in previous studies [13]. Under stoichiometric conditions, the limited radical pool suppresses the oxidation of NH_i to NO while simultaneously inhibiting N_2O decomposition, thereby shifting the balance toward N_2O accumulation. In contrast, under fuel-rich conditions, despite the promoted generation of NH_i radicals, the substantially reduced NO and NO_2 concentrations limit the availability of key reactants for N_2O production. As the temperature increased to 1000 °C, the N_2O concentration became barely detectable over the entire equivalence ratio. This behavior can be attributed to the thermal instability of N_2O and its enhanced decomposition at elevated temperatures, primarily via third-body reaction (R12) as well as reactions with reactive radicals such as O and H (R13) [43,44]. Consistent with this interpretation, previous studies have also reported the conversion of N_2O and NO at elevated temperatures [45]. These observations suggest that N_2O preferentially formed under lean-to-stoichiometric combustion conditions and was progressively suppressed with increasing temperatures, consistent with the findings reported in Ref. [26]. For partially cracked $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures ($\gamma = 0.3$), the variation trend of N_2O with

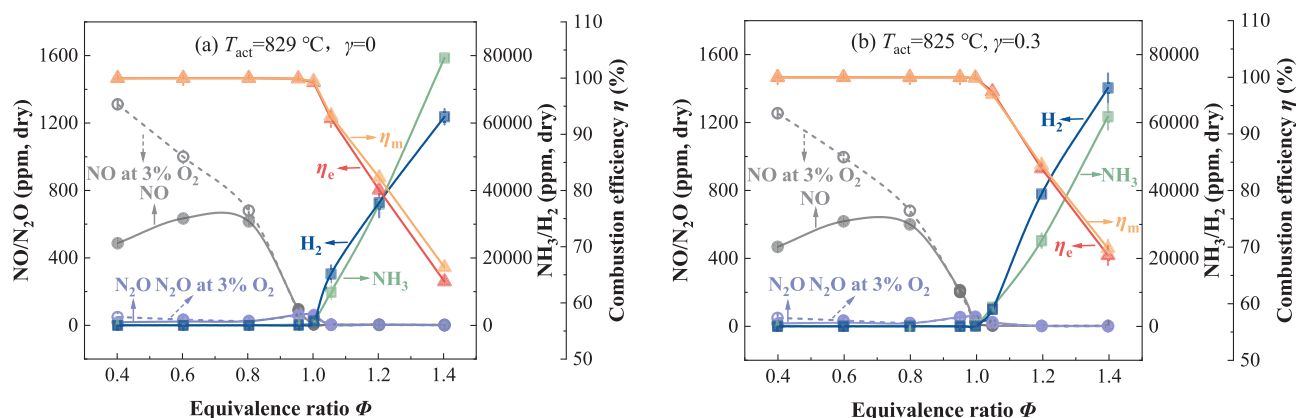


Fig. 4. Effects of equivalence ratio on species distributions and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) at a preset temperature of 800 °C.

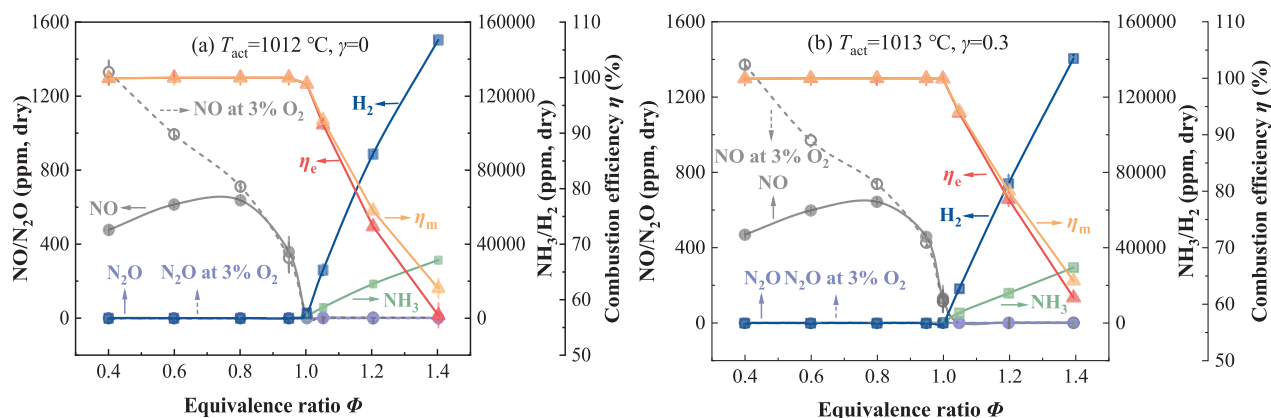


Fig. 5. Effects of equivalence ratio on speciation data and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) at a preset temperature of 1000 °C.

equivalence ratio was also essentially identical to that observed for pure NH_3 .



The NH_3 and H_2 concentrations exhibited similar trends, remaining nearly negligible under fuel-lean conditions ($\phi < 1$) and increasing sharply with rising equivalence ratio under fuel-rich conditions. For pure ammonia, NH_3 served as the sole hydrogen source, indicating that the detected H_2 originated primarily from the thermal decomposition of NH_3 , consistent with previous observations [20,27]. In contrast, for $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures, the detected H_2 might originate from both NH_3 decomposition and the externally supplied H_2 , making it difficult to decouple their individual contributions. As the temperature increased from 800 °C to 1000 °C, the H_2 concentration increased markedly while the NH_3 concentration decreased, indicating that NH_3 decomposition into H_2 was strongly temperature dependent [46]. Consequently, both η_m and η_e remained close to 100% under fuel-lean conditions but decreased substantially with increasing equivalence ratio under fuel-rich conditions. Notably, η_m coincided with η_e under fuel-lean conditions, whereas η_m became higher than η_e under fuel-rich conditions, with the discrepancy becoming more pronounced at elevated equivalence ratios. This discrepancy arose from the different sensitivities of these two efficiencies to the residual species. During NH_3 decomposition, hydrogen atoms were redistributed from NH_3 to H_2 while the total number of hydrogen atoms in the exhaust remained unchanged. However, H_2 retained more fuel chemical energy per hydrogen atom than NH_3 as indicated by their volumetric energy densities (NH_3 : 14.34 MJ/m³; H_2 : 10.68 MJ/m³). Consequently, η_m exceeded η_e under fuel-rich conditions. The discrepancy between η_m and η_e became more pronounced with increasing temperature, as elevated temperatures promoted NH_3 decomposition into H_2 , increasing the fraction of unreacted H_2 . In contrast, for $\text{NH}_3/\text{H}_2/\text{N}_2$ blends, externally supplied H_2 moderated NH_3 decomposition pathways, thereby mitigating the discrepancy.

Overall, near stoichiometric combustion was found to effectively balance combustion efficiency and NOx emission control. Nevertheless, the potential formation of N_2O , which exhibited a 100-year global warming potential approximately 273 times that of CO_2 [47], should not be overlooked. Accordingly, the following sections focus on near-stoichiometric conditions, with ultra-lean ($\phi = 0.4$) and ultra-rich ($\phi = 1.4$) conditions included for comparison. Due to the limited resolution and stability of the flow meters, as well as the high sensitivity of species distributions to small variations in air flow rates near $\phi \approx 1$, strictly stoichiometric conditions are difficult to precisely control, and multiple

experimental runs are typically required. Therefore, slightly lean ($\phi = 0.95$) and slightly rich ($\phi = 1.05$) conditions were adopted in the subsequent sections to ensure better experimental consistency and repeatability.

3.3. Effects of temperature

Beyond equivalence ratio, temperature fundamentally governs $\text{NH}_3/\text{H}_2/\text{N}_2$ combustion by modulating radical-mediated chemical kinetics and reaction pathways. In this section, the influence of temperature on species evolution and combustion efficiency was systematically investigated under the selected equivalence ratios, and the corresponding results are presented in Figs. 6–9.

As shown in Fig. 6, under ultra-lean conditions ($\phi = 0.4$), pure ammonia combustion at 700 °C was kinetically limited, resulting in substantial residual NH_3 and a very low combustion efficiency of 11.56%. In contrast, only small amounts of H_2 were detected in the exhaust, highlighting the temperature dependence of NH_3 decomposition. Notably, significant N_2O was observed under this condition, consistent with previous study reporting that factors weakening combustion stability substantially promoted N_2O formation [23]. Increasing the temperature [42] or employing partially cracked $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures [36,40] markedly enhanced the generation of reactive radicals (e. g., H, O and OH), with the latter attributed to H_2 -mediated elementary reactions (R14–R15). The increased radical pool consequently enhanced NH_3 conversion and reduced unreacted NH_3 in the exhaust. For both pure ammonia and $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures, the NO concentration in the exhaust exhibited a non-monotonic dependence on temperature, increasing initially, peaking at approximately 900 °C, and subsequently decreasing at elevated temperatures. The initial increase in NO formation was attributed to the enhanced generation of reactive radicals, which made the NOx formation pathways more competitive [28]. However, with further temperature increase, intensified NH_3 pre-decomposition upstream of the isothermal zone reduced the availability of NH_i intermediates for NO formation, suppressing NO production. In contrast, the outlet N_2O concentration decreased monotonically with increasing temperatures and became negligible above 900 °C, reflecting the thermal instability of N_2O , as discussed above.



For both pure NH_3 and $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures, the residual NH_3 concentration in the exhaust decreased sharply as the temperature increased from 700 °C to 800 °C and subsequently fluctuated around 100 ppm at higher temperatures. The H_2 concentration followed a similar decreasing trend and approached nearly zero once the

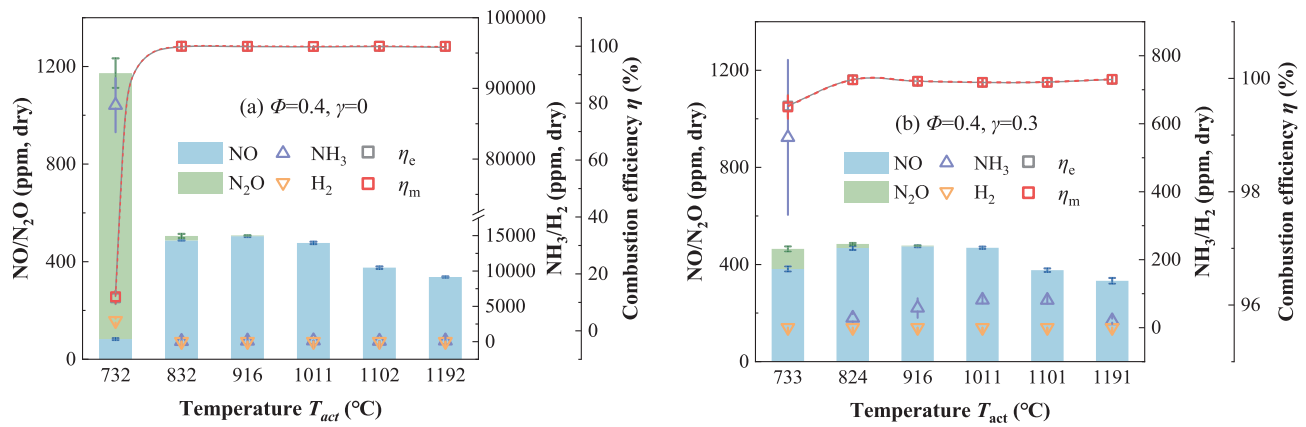


Fig. 6. Effects of temperature on speciation data and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) under $\phi = 0.4$.

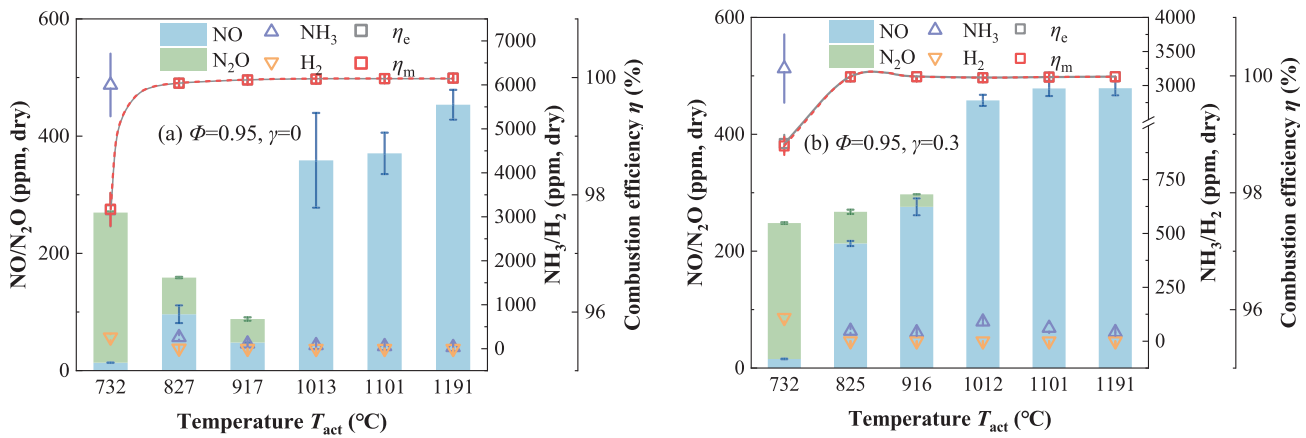


Fig. 7. Effects of temperature on speciation data and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) under $\phi = 0.95$.

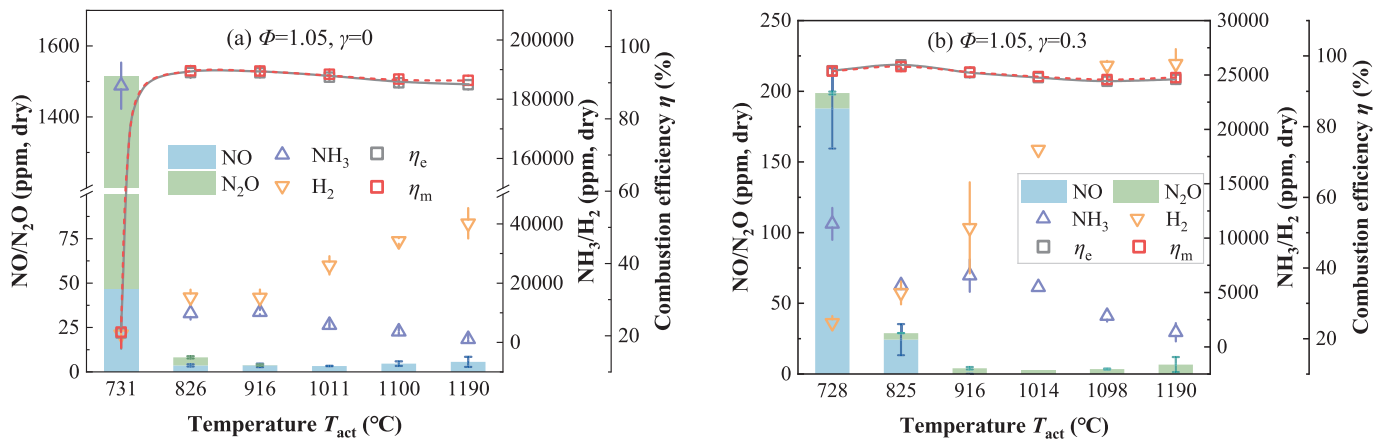


Fig. 8. Effects of temperature on speciation data and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) under $\phi = 1.05$.

temperature exceeded 800 °C. Consequently, both the energy-based (η_e) and mass-based (η_m) combustion efficiencies increased with temperature, reaching approximately 100% above 800 °C. Notably, η_e overlapped with η_m under this equivalence ratio. According to Eq. (9), the difference between η_m and η_e is related to the inlet and outlet NH_3 and H_2 flow rates. For pure NH_3 combustion (Fig. 6(a)), the inlet H_2 flow rate is zero, and the difference is therefore mainly determined by the outlet H_2 flow rate, which remained very small (thousands of ppm) under the present conditions. For $\text{NH}_3/\text{H}_2/\text{N}_2$ combustion (Fig. 6(b)), the concentrations of residual NH_3 and H_2 in the exhaust gas were both low (on

the order of hundreds of ppm). Therefore, only a negligible difference between η_m and η_e was observed.

$$(\eta_m - \eta_e) \propto [\dot{V}_{\text{out}} \cdot x_{\text{H}_2} \cdot (V_{\text{f, NH}_3} - V_{\text{i, NH}_3}) / (t_{\text{f}} - t_{\text{i}}) - V_{\text{o, NH}_3} / (t_{\text{f}} - t_{\text{i}}) \cdot (V_{\text{f, H}_2} - V_{\text{i, H}_2}) / (t_{\text{f}} - t_{\text{i}})] \quad (9)$$

Compared with ultra-lean condition, pure ammonia combustion at 700 °C under slightly lean condition ($\phi = 0.95$) exhibited markedly lower NH_3 and H_2 concentrations in the exhaust, accompanied by a high

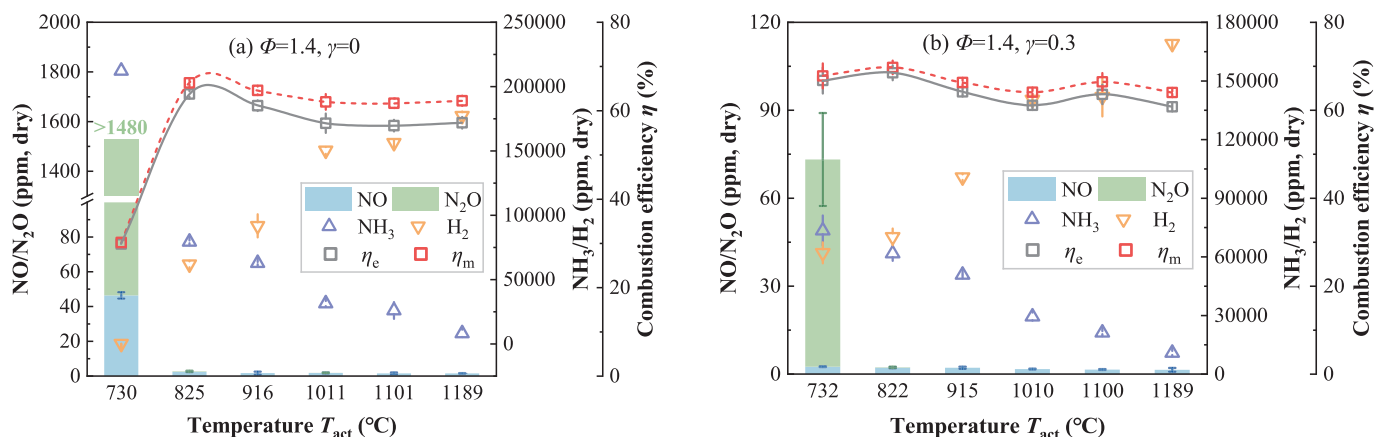


Fig. 9. Effects of temperature on speciation data and combustion efficiency for (a) pure ammonia ($\gamma = 0$) and (b) $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ($\gamma = 0.3$) under $\phi = 1.4$.

combustion efficiency of approximately 98%, as shown in Fig. 7. At $\phi = 0.4$, although the abundant O_2 facilitated radical formation, the severe dilution of nitrogen-containing species caused by excess air significantly limited NH_3 conversion. In contrast, at $\phi = 0.95$, the slight excess of O_2 sustained an adequate radical pool without pronounced dilution, thereby promoting NH_3 conversion. With increasing temperature, NO exhibited a trend distinct from that observed under ultra-lean conditions, increasing continuously and gradually leveling off once the temperature exceeded 1000 °C. Under this near-stoichiometric condition, the oxygen supply was only slightly above the stoichiometric requirement. As temperature increased, the production of reactive radicals such as OH and O was enhanced, which promoted the oxidation of nitrogen-containing intermediates and consequently strengthened the NO formation pathways [28]. In contrast, the temperature-dependent behaviors of N_2O , residual NH_3 , H_2 , and combustion efficiency remained similar to those under ultra-lean conditions. However, under slightly lean conditions, the temperature window for N_2O formation extended toward higher temperatures, which was likely attributed to the suppression of radical-mediated N_2O consumption pathways (R12-R13) due to the reduced abundance of reactive radicals.

Under both slightly rich ($\phi = 1.05$) and ultra-rich conditions ($\phi = 1.4$), pure ammonia combustion at 700 °C was kinetically limited, as evidenced by substantial NH_3 slip and extremely low combustion efficiency (Figs. 8 and 9). At 700 °C, the reduced O_2 availability under rich conditions was insufficient to sustain radical replenishment, thereby preventing effective NH_3 conversion. In addition, pronounced N_2O formation was observed, with the N_2O concentration exceeding the detection limit of the gas analyzer at $\phi = 1.4$. For pure ammonia, NO and N_2O formation predominated at 700 °C but became barely detectable at temperatures above 800 °C. This behavior was attributed to the increasing dominance of DeNOx pathways at elevated temperatures, where excess nitrogen-containing radicals preferentially reduced NO [48]. For $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures, NO formation at 700–800 °C was promoted under slightly rich conditions but was suppressed at 700 °C under ultra-rich conditions. This contrast reflected the competition between nitrogen-containing radical oxidation pathways and NH_3 -mediated DeNOx reactions. Under slightly rich conditions, the oxygen availability was only moderately lower than that under stoichiometric conditions, and H_2 addition enhanced radical generation and NH_3 conversion [49], making fuel-nitrogen oxidation pathways more competitive. In contrast, under ultra-rich conditions, excess NH_3 radicals and severe oxygen deficiency shifted the reaction network toward NH_3 -mediated NO reduction, thereby suppressing NO formation.

With increasing temperature, the residual NH_3 concentration decreased progressively, while the outlet H_2 concentration increased. The pronounced reduction in residual NH_3 from 700 °C to 800 °C was primarily attributed to enhanced NH_3 combustion, as evidenced by the

concurrent increase in combustion efficiency and the relatively modest rise in H_2 formation. At higher temperatures (>800 °C), the further decrease in residual NH_3 was mainly driven by intensified NH_3 decomposition into N_2 and H_2 , as indicated by the progressive reduction in energy-based combustion efficiency. As NH_3 decomposition into N_2 and H_2 is an endothermic process ($\Delta H_{298} = 46 \text{ kJ/mol}$) [50], it increased the total chemical energy retained in the outlet gases, thereby reducing η_e with increasing temperature. In addition, the decrease in η_m also contributed to the reduction in η_e . At elevated temperatures, the shortened residence time likely led to increased slip of unreacted fuels (e.g., NH_3 and H_2), resulting in greater chemical energy losses and consequently a lower η_e . Notably, under slightly rich conditions, η_m exceeded η_e only at high temperatures, whereas under ultra-rich conditions, η_m remained consistently higher than η_e . According to Eq. (9), intensified NH_3 decomposition under these conditions increased the outlet H_2 flow rate while decreasing the outlet NH_3 flow rates, thereby enlarging the difference between η_m and η_e .

To further elucidate the NH_3 conversion mechanisms, the individual contributions of different NH_3 consumption pathways under fuel-rich conditions were quantified and are presented as stacked bar charts in Fig. 10. Four principal NH_3 conversion channels were considered for pure NH_3 combustion: oxidation to N_2 (P1: $4\text{NH}_3 + 3\text{O}_2 \rightarrow 2\text{N}_2 + 6\text{H}_2\text{O}$), thermal decomposition (P2: $2\text{NH}_3 \rightarrow \text{N}_2 + 3\text{H}_2$), oxidation to NOx (P3: $\text{NH}_3 + \text{O}_2 \rightarrow \text{NO}_x + \text{H}_2\text{O}$), and unreacted NH_3 (P4). The contribution of each channel was quantified based on elemental balances. Since NH_3 is the sole hydrogen-containing species in the feed, the conversion via P2 was inferred from the measured H_2 concentration. Given that thermal NO formation is negligible within the investigated temperature range, NOx was assumed to originate predominantly from fuel-N under present conditions. Accordingly, the contribution of P3 was evaluated from the fraction of nitrogen atoms present in NO and N_2O relative to the total nitrogen introduced by NH_3 . The fraction of unreacted NH_3 (P4) was directly obtained from the measured outlet NH_3 concentration. Finally, the contribution of P1 was determined by difference to satisfy the overall nitrogen balance. A similar pathway analysis was not performed for $\text{NH}_3/\text{H}_2/\text{N}_2$ mixtures because the H_2 produced from NH_3 decomposition and the externally supplied H_2 were highly coupled and could not be unambiguously distinguished based on the experimentally measured species concentrations alone. Therefore, the elemental balance approach employed for pure NH_3 combustion could not be directly extended to the ternary system without additional assumptions. As shown in Fig. 10, NH_3 conversion was predominantly governed by P1 across the investigated temperature range, except at 700 °C, while the contribution of P3 remained negligible. At temperatures above 800 °C, the contributions of P1 and P4 decreased progressively with increasing temperature, whereas the contribution of P2 gradually increased, revealing intensified NH_3 thermal decomposition at elevated

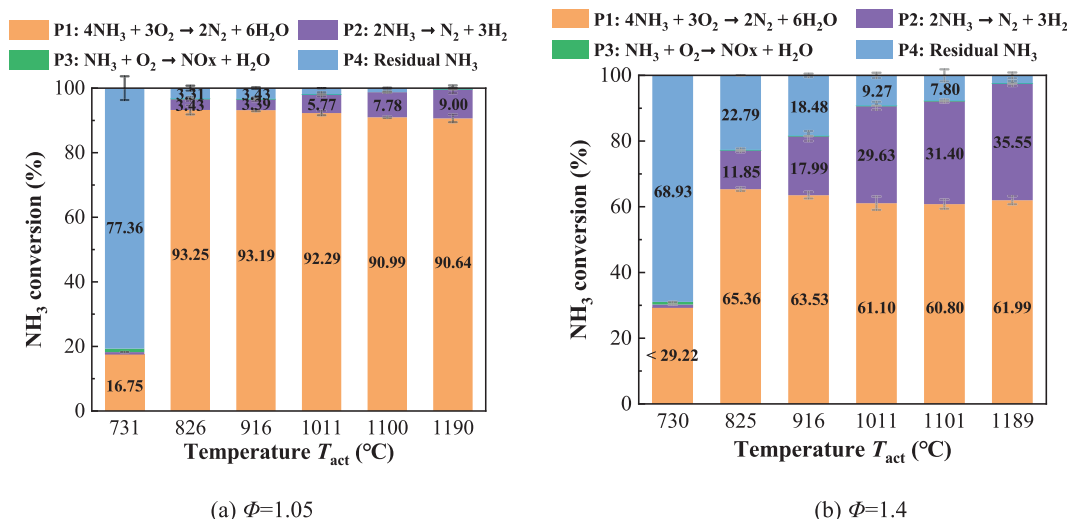


Fig. 10. Individual contributions of different NH₃ conversion pathways.

temperatures. Notably, the fraction of NH₃ converted via P1 exhibited good agreement with the mass-based combustion efficiency (η_m). This observation was consistent with the definition of η_m as the fraction of NH₃ converted to H₂O through oxidative pathways (P1 and P3), with P1 dominating due to the negligible contribution of P3.

To summarize, NH₃ combustion was kinetically limited at 700 °C under both ultra-lean and rich conditions, whereas it proceeded much more effectively under slightly lean conditions. Increasing temperature and hydrogen blending promoted the generation of reactive radicals, thereby facilitating NH₃ conversion. N₂O was predominantly formed under kinetically limited conditions and gradually diminished with increasing temperature owing to its thermal instability. The temperature dependence of NO formation was strongly influenced by the equivalence ratio. Under ultra-lean conditions, NO exhibited a non-monotonic trend with temperature due to the competition between NO formation pathways and enhanced NH₃ pre-decomposition. Under slightly lean conditions, NO increased monotonically with temperature as a result of enhanced radical production. In contrast, under rich conditions, NO formation predominated at 700 °C but became nearly negligible at elevated temperatures, owing to the temperature-dependent DeNO_x mechanism. With increasing temperature, the residual NH₃ concentration decreased progressively. Meanwhile, the H₂ concentration

decreased under lean conditions but increased steadily under rich conditions. These contrasting trends led to distinct combustion efficiency behaviors: under lean conditions, combustion efficiency increased with temperature and approached a plateau at elevated temperatures, whereas under rich conditions it slightly declined at high temperatures due to shortened residence time and enhanced NH₃ thermal decomposition. Notably, under rich conditions at elevated temperatures, the mass-based combustion efficiency remained higher than the energy-based combustion efficiency.

3.4. Effects of NH₃ decomposition ratio

Partial ammonia cracking provides a practical route to generate NH₃/H₂/N₂ mixtures with controllable compositions. Varying the NH₃ decomposition ratio directly alters the relative proportions of NH₃, H₂, and N₂ in the fuel, thereby modifying radical chemistry and potentially influencing combustion performance. Therefore, this section systematically investigates the effects of NH₃ decomposition ratio on speciation distributions and combustion efficiency under four typical equivalence ratios at 800 and 1000 °C. These temperatures were selected to minimize kinetic limitations on NH₃ conversion, thereby reducing the influence of reaction rate constraints on the observed behavior. In addition, 1000 °C

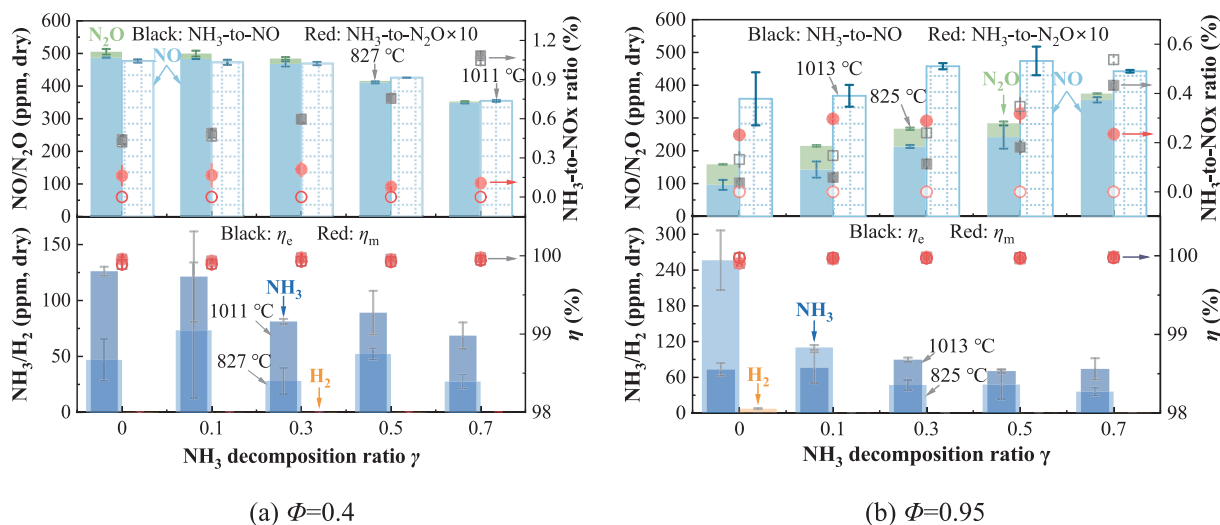


Fig. 11. Effects of NH₃ decomposition ratio on speciation data and combustion efficiency under lean conditions. Filled and empty symbols correspond to preset temperatures of 800 °C and 1000 °C, respectively.

was selected because secondary thermal decomposition of unreacted NH_3 is more pronounced than at 800 °C, as identified in Section 3.3.

The evolution of speciation distribution and combustion efficiency with NH_3 decomposition ratio under lean conditions is illustrated in Fig. 11. At both 800 °C and 1000 °C, thermal NO formation is expected to be negligible due to relatively low combustion temperatures, and NO formation is therefore mainly associated with fuel-NO pathways involving NH_i intermediates. Under ultra-lean conditions, the NO concentration decreased slightly as the NH_3 decomposition ratio increased from 0 to 0.3, followed by a pronounced reduction at higher decomposition ratios. This trend appeared inconsistent with previous studies, in which the NO concentration initially increased and subsequently decreased with increasing NH_3 decomposition ratio [28,51,52]. The non-monotonic trend reported in the literature has been attributed to the competition between the chemical and the dilution effect associated with NH_3 partial decomposition. Specifically, H_2 addition enhanced radical generation (R14-R15), thereby promoting the oxidation of nitrogen intermediates to NO via reactions R1-R5, particularly at lower NH_3 decomposition ratios [53]. In contrast, NH_3 partial cracking decreased the inlet NH_3 concentration, limiting the availability of nitrogen intermediates for NO formation, which became increasingly important at high NH_3 decomposition ratios. However, Ditaranto et al. [23] further highlighted the role of adiabatic temperature increase induced by H_2 in the observed non-monotonic NO behavior. When the adiabatic temperature was maintained constant in their calculation, the NO concentration increased monotonically with increasing NH_3 concentration in the fuel. Since the reactor temperature in the present study was externally controlled, the contribution of adiabatic temperature rise was considered negligible. Under such conditions, radical production under ultra-lean operation was already rapid, reducing the relative contribution of H_2 -mediated radical enhancement compared with the dilution effect and thereby leading to an overall decrease in the measured NO concentration. This trend is qualitatively consistent with the constant-temperature predictions reported by Ditaranto et al. [23]. Moreover, the gradual decrease in O_2 concentration, as calculated from the inlet flow rates, may have further suppressed NO formation. To decouple the dilution effect, the NH_3 -to-NO conversion ratio was evaluated. As shown in Fig. 11(a), this ratio increased monotonically with increasing decomposition ratio, indicating that on a per- NH_3 basis, H_2 addition indeed promoted NO formation despite the apparent decrease in measured NO concentration.

At 800 °C, the N_2O concentration exhibited a trend similar to that of NO, decreasing slightly as γ increased from 0 to 0.3 and then dropping sharply to nearly zero at higher decomposition ratios. In contrast, N_2O was barely detectable at 1000 °C owing to its thermal instability. Partial cracking of NH_3 into $\text{NH}_3/\text{H}_2/\text{N}_2$ mixture also exerted competing effects on N_2O formation. As discussed above, N_2O was mainly formed via the $\text{NH} + \text{NO}$ (R9) and $\text{NH}_2 + \text{NO}_2$ (R11) reactions. On the one hand, H_2 addition enhanced radical generation and promoted the formation of NH_i intermediates, thereby favoring N_2O formation. On the other hand, partial cracking of NH_3 diluted the inlet NH_3 concentration, reducing the availability of nitrogen intermediates involved in N_2O formation. Meanwhile, the enhanced radical pool associated with H_2 addition accelerated the conversion of N_2O to N_2 (R13). At higher decomposition ratios, the dilution effect and the enhanced N_2O conversion became dominant, leading to a pronounced reduction in N_2O concentration. In contrast, the NH_3 -to- N_2O conversion ratio exhibited a non-monotonic dependence on the NH_3 decomposition ratio, indicating the coexistence of these competing effects.

Owing to the presence of excess air, the outlet H_2 concentration remained nearly zero, whereas NH_3 fluctuated at relatively low levels across all NH_3 decomposition ratios. This difference was likely attributable to the higher diffusivity and reactivity of H_2 , which enabled its rapid oxidation. Consequently, both the mass-based and energy-based combustion efficiencies remained close to 100% at both temperatures across all decomposition ratios, exhibiting nearly identical values.

In contrast, under slightly lean conditions, the NO concentration increased progressively with increasing NH_3 decomposition ratios at 800 °C, whereas at 1000 °C a decrease in NO concentration was observed when the decomposition ratio was further increased to 0.7. At $\phi = 0.95$, the supplied O_2 was only marginally higher than the stoichiometric requirement, and the oxidation of nitrogen-containing intermediates was therefore limited by the availability of reactive radicals (OH , O , and H). Although inlet NH_3 and O_2 concentration and residence time decreased with increasing NH_3 decomposition ratio, H_2 -mediated reactions markedly enhanced the production of reactive radical. At 800 °C, the promoting effects of H_2 addition outweighed these inhibitory effects, leading to progressively higher NO concentrations. At 1000 °C, however, reactive radical generation was already sufficiently rapid, making the promoting effect of H_2 less pronounced. Consequently, the combined inhibitory effects associated with lower NH_i and O_2 concentrations together with the shortened residence time at higher NH_3 decomposition ratios outweighed the promoting effect, resulting in a decrease in NO concentration. Similar to the ultra-lean case, the NH_3 -to-NO ratio increased monotonically with increasing NH_3 decomposition ratio.

At 800 °C, both the measured N_2O concentration and NH_3 -to- N_2O ratio exhibited a non-monotonic dependence on the NH_3 decomposition ratio, initially increasing and subsequently decreasing, reflecting the competing effects associated with NH_3 partial decomposition. In contrast, at 1000 °C, N_2O remained barely detectable across all NH_3 decomposition ratios. At 800 °C, the residual NH_3 concentration decreased progressively with increasing NH_3 decomposition ratio and eventually approached a plateau, whereas only trace amounts of H_2 were observed in the exhaust during pure ammonia combustion. By contrast, at 1000 °C, H_2 remained negligible over the entire range of decomposition ratios, whereas NH_3 fluctuated at relatively lower levels. These distinct behaviors could be attributed to the limited availability of reactive radicals under slightly lean conditions at lower temperatures, whereas elevated temperatures substantially enhanced radical generation. As a result, the combustion efficiency of pure ammonia at 1000 °C was slightly higher than that at 800 °C. Upon partial cracking, however, the combustion efficiencies at both temperatures became nearly identical ($\sim 100\%$), with η_e overlapping with η_m across all NH_3 decomposition ratios.

The evolution of species concentrations and combustion efficiency with NH_3 decomposition ratio under rich conditions is illustrated in Fig. 12. Under slightly rich conditions ($\phi = 1.05$), NO concentrations remained low across the entire range of NH_3 decomposition ratios at both temperatures owing to the predominance of DeNOx pathways, although a slight increase was observed at $\gamma = 0.7$ at 800 °C. N_2O remained negligible at 1000 °C; however, at 800 °C, both the measured N_2O and NH_3 -to- N_2O ratio increased markedly with increasing NH_3 decomposition ratio. In contrast, under ultra-rich conditions ($\phi = 1.4$), both NO and N_2O concentrations remained negligible regardless of temperature or decomposition ratio. The distinct N_2O formation behaviors observed at 800 °C between $\phi = 1.05$ and $\phi = 1.4$ were primarily governed by oxygen availability. Under slightly rich conditions, the marginal oxygen deficiency rendered DeNOx pathways less competitive than under ultra-rich conditions. In this regime, increasing γ promoted NH_3 conversion and the formation of NH_i intermediates, thereby facilitating N_2O formation through $\text{NH} + \text{NO}$ and $\text{NH}_2 + \text{NO}_2$ reactions. Conversely, under ultra-rich conditions, the severely oxygen-deficient environment strongly favored NH_3 -to- N_2 conversion, while the limited availability of NO effectively suppressed N_2O formation.

Under slightly rich condition, the residual NH_3 concentration decreased monotonically with increasing γ at 800 °C. This behavior resulted from the combined effects of enhanced NH_3 conversion induced by H_2 addition and the dilution effect, in agreement with previous studies [50]. Meanwhile, the outlet H_2 concentration also decreased progressively with increasing γ . As discussed above, H_2 contains fewer hydrogen atoms per unit energy than NH_3 ; therefore, for the same

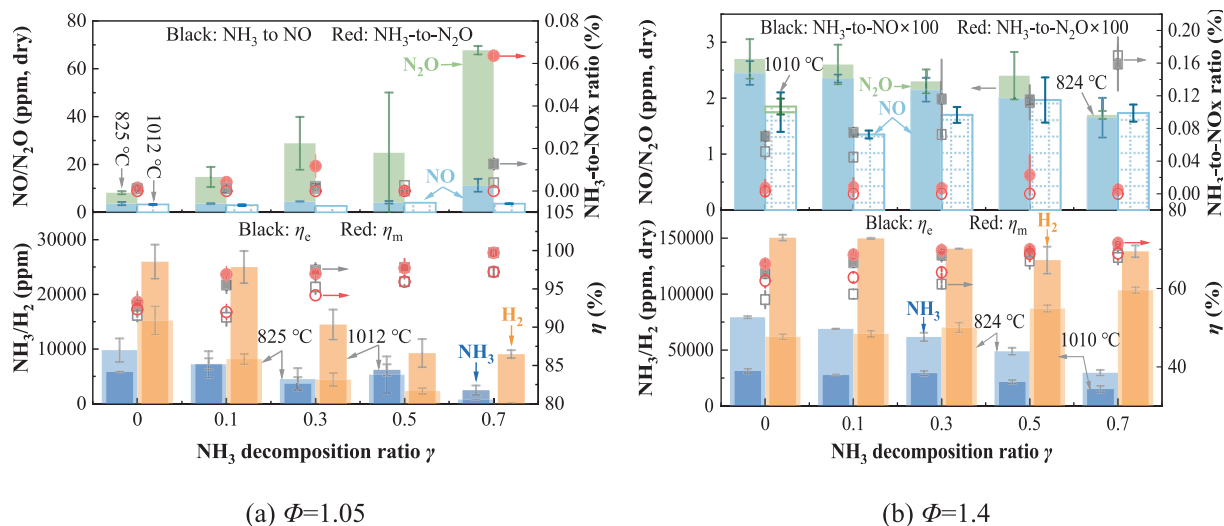


Fig. 12. Effects of NH_3 decomposition ratio on speciation data and combustion efficiency under rich conditions. Filled and empty symbols correspond to preset temperatures of 800 °C and 1000 °C, respectively.

energy input, increasing γ reduced the total amount of hydrogen atoms introduced into the system. Under slightly rich conditions, the marginal oxygen deficiency still allowed effective oxidation of hydrogen-containing species, leading to a gradual decrease in residual H_2 concentration with increasing γ . Accordingly, both the mass-based (η_m) and energy-based (η_e) combustion efficiencies increased steadily with increasing γ and gradually approached a plateau at higher γ . The initial increase could be attributed to the distinct reactivities of H_2 and NH_3 . On the one hand, H_2 addition promoted the generation of reactive radicals, which facilitated NH_3 conversion and thereby enhanced combustion efficiency. On the other hand, owing to its higher reactivity, H_2 preferentially consumed the available O_2 and competed with NH_3 oxidation. Since H_2 provided more energy per hydrogen atom than NH_3 , the preferential oxidation of H_2 contributed to a greater energy release, further increasing η_e . The appearance of a plateau indicated the existence of a theoretical upper limit. The theoretical upper limit of η_m can be derived based on the definition of equivalence ratio. The stoichiometric air requirement is determined by assuming complete conversion of all hydrogen in the fuel to H_2O , while nitrogen is assumed to form N_2 and does not consume oxygen. At $\Phi = 1.05$, the actual air supply corresponds to 1/1.05 of the stoichiometric air. Therefore, even if all supplied air is fully utilized for the oxidation of hydrogen to H_2O , the oxygen-limited condition restricts hydrogen conversion to a maximum fraction of 1/1.05. This corresponds to the theoretical upper limit of η_m , which is 95.24%. Notably, the plateau value of η_m slightly exceeded this limit, which was most likely attributable to the limited resolution of the flow meters and measurement uncertainties.

Similar to the observations at 800 °C, the residual NH_3 and H_2 concentrations generally decreased with increasing γ at 1000 °C, although minor fluctuations were observed in the NH_3 concentration. However, the outlet H_2 concentration remained higher than that at 800 °C, reflecting enhanced NH_3 decomposition at elevated temperatures. Moreover, both the energy-based (η_e) and mass-based (η_m) combustion efficiencies were lower at 1000 °C. As discussed above, the reduction in η_m at elevated temperatures was mainly attributed to the shortened residence time, whereas the decrease in η_e originated not only from the retention of hydrogen-containing species in the exhaust but also from the intensified NH_3 thermal decomposition.

Under ultra-rich conditions at 800 °C, the residual NH_3 concentration decreased with increasing γ , whereas the outlet H_2 increased monotonically with γ , showing a trend opposite to that observed under slightly rich conditions. Compared with $\Phi = 1.05$, the oxygen supply at $\Phi = 1.4$ was far below the stoichiometric requirement and was

insufficient to support effective oxidation of hydrogen-containing species. As a result, pronounced H_2 slip occurred, leading to a continuous increase in the outlet H_2 concentration with increasing γ . In contrast, at 1000 °C, both NH_3 and H_2 concentrations decreased with increasing γ . The distinct H_2 evolution behavior reflected enhanced reaction kinetics at elevated temperatures, under which hydrogen-containing species could be more effectively converted. Under these conditions, the reduced hydrogen atom input with increasing γ resulted in a monotonic decrease in the outlet H_2 concentration.

Both η_e and η_m initially increased and then reached a plateau at $\gamma \approx 0.5$, suggesting that further H_2 addition provided limited benefit due to the existence of a theoretical upper limit. At $\Phi = 1.4$, the theoretical upper limit of η_m is 71.43% ($=1/1.4$), which was in good agreement with the plateau value observed at 800 °C. Notably, under ultra-rich conditions, both η_m and η_e decreased with increasing temperature, and η_m remained consistently higher than η_e , as discussed above.

In summary, hydrogen blending resulted in a progressive decrease in both measured NO and N_2O concentrations under ultra-lean conditions, while the combustion efficiency remained close to 100%. Under slightly lean conditions, the NO concentration increased monotonically with increasing NH_3 decomposition ratio at 800 °C, whereas at 1000 °C it decreased at higher decomposition ratios. In contrast, N_2O exhibited a non-monotonic dependence on the NH_3 decomposition ratio. Hydrogen blending also significantly enhanced combustion efficiency at lower temperatures. Under slightly rich conditions, both NO and N_2O concentrations increased with increasing NH_3 decomposition ratio, accompanied by progressive decrease in residual NH_3 and H_2 . By contrast, under ultra-rich conditions, both NO and N_2O remained negligible. In this regime, the residual NH_3 concentration decreased continuously at both temperatures, while the outlet H_2 concentration increased with increasing decomposition ratio at 800 °C but decreased at 1000 °C.

3.5. Comprehensive evaluation of the combustion performance

As discussed above, the equivalence ratio, combustion temperature, and NH_3 decomposition ratio exerted coupled influence on NO and N_2O emissions as well as combustion efficiency, resulting in trade-offs among these performance indicators. To enable a comprehensive comparison of combustion performance under different conditions, radar charts were employed to jointly visualize NO and N_2O emissions together with combustion efficiencies. Only slightly lean and slightly rich conditions that achieved near-complete combustion were selected for further

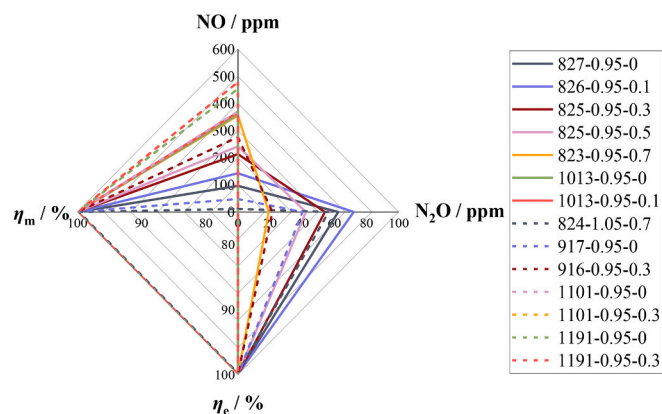


Fig. 13. Multi-criteria comparison of combustion performance under selected operating conditions.

comparison, as sufficient combustion efficiency and low NO_x emissions were fundamental requirements for practical power generation systems. As shown in Fig. 13, the numbers in the legend represent temperature, equivalence ratio, and NH₃ decomposition ratio in sequence. For example, 800–0.95–0 corresponds to pure ammonia combustion at 800 °C with an equivalence ratio of 0.95. The radar charts clearly illustrate that no single operating conditions simultaneously optimized all performance indicators, highlighting the trade-offs between NO and N₂O emissions and combustion efficiency.

Temperatures above 1000 °C effectively suppressed N₂O emissions to nearly zero while maintaining high combustion efficiencies (η_e and $\eta_m > 99.97\%$) under slightly lean conditions. However, these benefits were accompanied by substantially elevated NO concentrations (359–479 ppm), indicating that further increases in temperature offered limited advantages from an overall perspective. Therefore, although these conditions may be preferred when complete suppression of N₂O emissions is prioritized, there are generally unfavorable for achieving balanced NO_x control.

Under slightly lean conditions, pure NH₃ combustion at 900 °C provided a particularly favorable balance between NO and N₂O emissions while maintaining high combustion efficiencies ($\eta_e = 99.96\%$ and $\eta_m = 99.96\%$). Compared with pure NH₃ combustion at 800 °C, the NO and N₂O concentrations decreased from 96.2 and 62.5 ppm to 47.9 and 40.2 ppm, respectively, corresponding to reductions of approximately 50% and 36%, without sacrificing combustion efficiency. Although increasing the NH₃ decomposition ratio further reduced N₂O emissions, this benefit was accompanied by a progressive increase in NO concentration. Moreover, the promoting effect on NO formation associated with increasing decomposition ratio became more pronounced at 900 °C than at 800 °C. Consequently, at high decomposition ratios (i.e., $\gamma = 0.7$), the NO concentrations under slightly lean conditions increased to levels comparable to those observed at temperatures above 1000 °C, thereby diminishing the NO_x advantage associated with operation at 800–900 °C. These results suggest that the use of partially cracked NH₃ under slightly lean conditions should be carefully evaluated according to the relative importance assigned to NO and N₂O.

In contrast, under slightly rich conditions, the condition 800–1.05–0.7 exhibited a distinct advantage when minimizing NO emissions was prioritized. This condition achieved the lowest NO concentration among all selected conditions (11.3 ppm), although it was accompanied by moderately higher N₂O emissions (56.5 ppm) and a slight reduction in combustion efficiencies ($\eta_e = 99.74\%$ and $\eta_m = 99.72\%$). Therefore, this operating condition may be particularly attractive for applications subject to stringent NO emission requirements.

To summarize, no single operating condition simultaneously optimized all performance indicators. 800–900 °C appeared to be more

suitable for achieving a balance between combustion efficiency and NO_x emissions. Although temperatures above 1000 °C effectively eliminated N₂O emissions, this benefit was accompanied by substantially increased NO formation. Under slightly lean conditions, pure NH₃ combustion at both 800 and 900 °C achieved relatively low NO and N₂O emissions while maintaining high combustion efficiencies, with 900–0.95–0 exhibiting comparatively better overall performance. Increasing the NH₃ decomposition ratio further reduced N₂O emissions but simultaneously promoted NO formation, with the NO-promoting effect becoming more pronounced at higher temperatures. In contrast, under slightly rich conditions, the condition 800–1.05–0.7 achieved the lowest NO concentration among all selected conditions and may therefore be preferred when ultra-low NO emissions are prioritized. Therefore, the preferred operating condition ultimately depends on the relative importance assigned to NO mitigation, N₂O suppression, and combustion efficiency.

4. Conclusions

In this study, the evolution of gaseous species and combustion efficiency during NH₃/H₂/N₂ combustion was systematically investigated over a temperature range of 700–1200 °C, covering a wide range of equivalence ratio (0.4–1.4) and NH₃ decomposition ratio (0–0.7). The reliability of experimental data was confirmed by nitrogen balance analysis (96.7%–110.0%). The key findings are summarized as follows.

- 1) NO formation dominates under lean conditions, whereas NH₃ and H₂ are favored under rich conditions. N₂O is mainly produced at 700–800 °C under lean-to-stoichiometric conditions, peaking near stoichiometry. Near-stoichiometric combustion offers a favorable trade-off between combustion efficiency and NO emissions.
- 2) Kinetic limitations in NH₃ combustion at 700 °C under ultra-lean and rich conditions led to significant N₂O formation and NH₃ slip, but was alleviated with increasing temperature. NO shows a non-monotonic temperature dependence at $\Phi = 0.4$ but increases monotonically at $\Phi = 0.95$, whereas N₂O generally decreases with temperature due to its thermal instability. Under lean conditions, NH₃ and H₂ decrease with temperature, whereas under rich conditions NH₃ decreases but H₂ increases. Accordingly, increasing temperature improves lean combustion efficiency below 800 °C but deteriorates rich combustion efficiency at elevated temperatures.
- 3) NH₃ decomposition ratio governs NO and N₂O formation through competing effects between enhanced radical generation and the reduced availability of nitrogen-containing intermediates. NH₃ concentration generally decreases with NH₃ decomposition ratio due to reduced NH₃ feed rate and enhanced conversion, with the largest reductions observed at 700 °C and under rich conditions. The dependence of H₂ on NH₃ decomposition ratio is governed by competing effects between increased H₂ feed rate and reduced overall H atom input. Hydrogen blending effectively enhanced combustion efficiency, with the most pronounced improvements observed at 700 °C and under rich conditions.
- 4) Temperatures of 800–900 °C favors balanced combustion performance, particularly for pure NH₃ combustion at $\Phi = 0.95$, whereas temperatures above 1000 °C diminishes N₂O emissions at the expense of substantial NO formation. At $\Phi = 0.95$, increasing NH₃ decomposition ratio reduces N₂O emissions but significantly promotes NO formation, while at $\Phi = 1.05$, a high decomposition ratio ($\gamma = 0.7$) favors lower NO emissions without significantly sacrificing combustion efficiency at 800 °C. Overall, the preferred operating condition ultimately depends on the targeted optimization objective.

This work provides guidance for optimizing NH₃/H₂/N₂ combustion in practical combustion systems and contributes to the decarbonization of the energy sector. However, the present study is primarily based on experimental investigations. Future work will focus on constructing

appropriate chemical reaction networks (CRNs) and refining detailed chemical kinetic mechanisms to elucidate the underlying mechanisms, together with validation of the findings in a larger-scale boiler currently under development.

CRedit authorship contribution statement

Zhixiang Zhu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Keisuke Shibata:** Investigation, Data curation. **Nobusuke Kobayashi:** Writing – review & editing, Supervision, Resources, Project administration. **Akira Suami:** Resources. **Tingting Gu:** Resources. **Ryo Yoshiie:** Project administration. **Shinji Kambara:** Project administration, Funding acquisition. **Xiaojiang Wu:** Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT in order to improve the clarity and readability of the language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2026.140640>.

Data availability

Data will be made available on request.

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